

**B.E. INFORMATION TECHNOLOGY FOURTH YEAR FIRST SEMESTER EXAM
2023**

DISTRIBUTED COMPUTING

Time: Three Hours

Full Marks: 100

- ✓A. (CO1) Consider a Complete Graph of 5 nodes. Describe a Leader Election Algorithm for it considering synchronous rounds. Count the No. of rounds. (15+8)
(CO2). What is global and local clock? Explain it for a graph network. Using Candy Lamport Algorithm how global snapshot is taken for a distributed system. (6+3+10) ✓✓

OR

(CO2) Explain Local and Global Timing in Distributed Algorithm. How Asynchrony is handled in distributed algorithm. Describe an application of Snapshot recording Algorithm. (5+5+6)

(CO1) Develop and analysis a distributed algorithm for Leader Election in a ring network of n nodes. Give a brief outline of correctness of your algorithm. Briefly explain Luby's algorithm for computing MIS in synchronous network. (10+6+10)

- B. (CO3) Write the Applications of Mutual Exclusion Methods. How progress is assured in mutual exclusion algorithm? What are the performance metrics of mutual exclusion? (7+7+7)

OR

- ✓(CO3) Explain Different Types of Asynchronous Shared Memory in brief. Describe a mutual exclusion algorithm. How progress is assured in asynchronous distributed system? (6+8+7) ✓
e. (CO4) Describe Kshemkalyani Algorithm for dead lock detection. Run the algorithm for 4 nodes with all having degree 2. Analyze the algorithm. (10+5+7) 2 2

OR

(CO4) What are the assumptions of Kshemkalyani and Singhal Deadlock detection algorithm? Explain the algorithm in brief. How termination is guaranteed. (5+10+7)

D. (CO5 & CO6) Choose the correct Option. (1.25 X 12)

1. _____ of the distributed file system are dispersed among various machines of distributed system.
a) Clients
b) Servers
c) Storage devices
d) All of the mentioned
2. Which one of the following hides the location where in the network the file is stored?
a) transparent distributed file system
b) hidden distributed file system
c) escaped distribution file system
d) spy distributed file system
3. In a distributed file system, when a file's physical storage location changes _____
a) file name need to be changed
b) file name need not to be changed
c) file's host name need to be changed
d) file's local name need to be changed
4. In a distributed file system, _____ is mapping between logical and physical objects.
a) client interfacing
b) naming
c) migration
d) heterogeneity
5. In a distributed file system, a file is uniquely identified by _____
a) host name
b) local name
c) the combination of host name and local name
d) none of the mentioned
6. In distributed file system, file name does not reveal the file's _____
a) local name
b) physical storage location
c) both local name and physical storage location
d) none of the mentioned
7. Which one of the following is a distributed file system?
a) andrew file system
b) network file system
c) novel network
d) all of the mentioned

8. What are the characteristics of tightly coupled system?
- i) Same clock, usually shared memory
 - ii) Communication is via this shared memory
 - iii) Multiprocessors
 - iv) Different clock
- a) i
b) i, ii and iii
c) ii and iii
d) i, iii and iv
9. What are the characteristics of mutual exclusion using centralized approach?
- a) One processor as coordinator which handles all requests
 - b) It requires request, reply and release per critical section entry
 - c) The method is free from starvation
 - d) All of the mentioned
10. The file once created can not be changed is called _____
- a) immutable file
 - b) mutex file
 - c) mutable file
 - d) none of the mentioned
11. _____ is not possible in distributed file system.
- a) File replication
 - b) Migration
 - c) Client interface
 - d) Remote access
12. In a distributed file system, _____ is mapping between logical and physical objects.
- a) client interfacing
 - b) naming
 - c) migration
 - d) heterogeneity